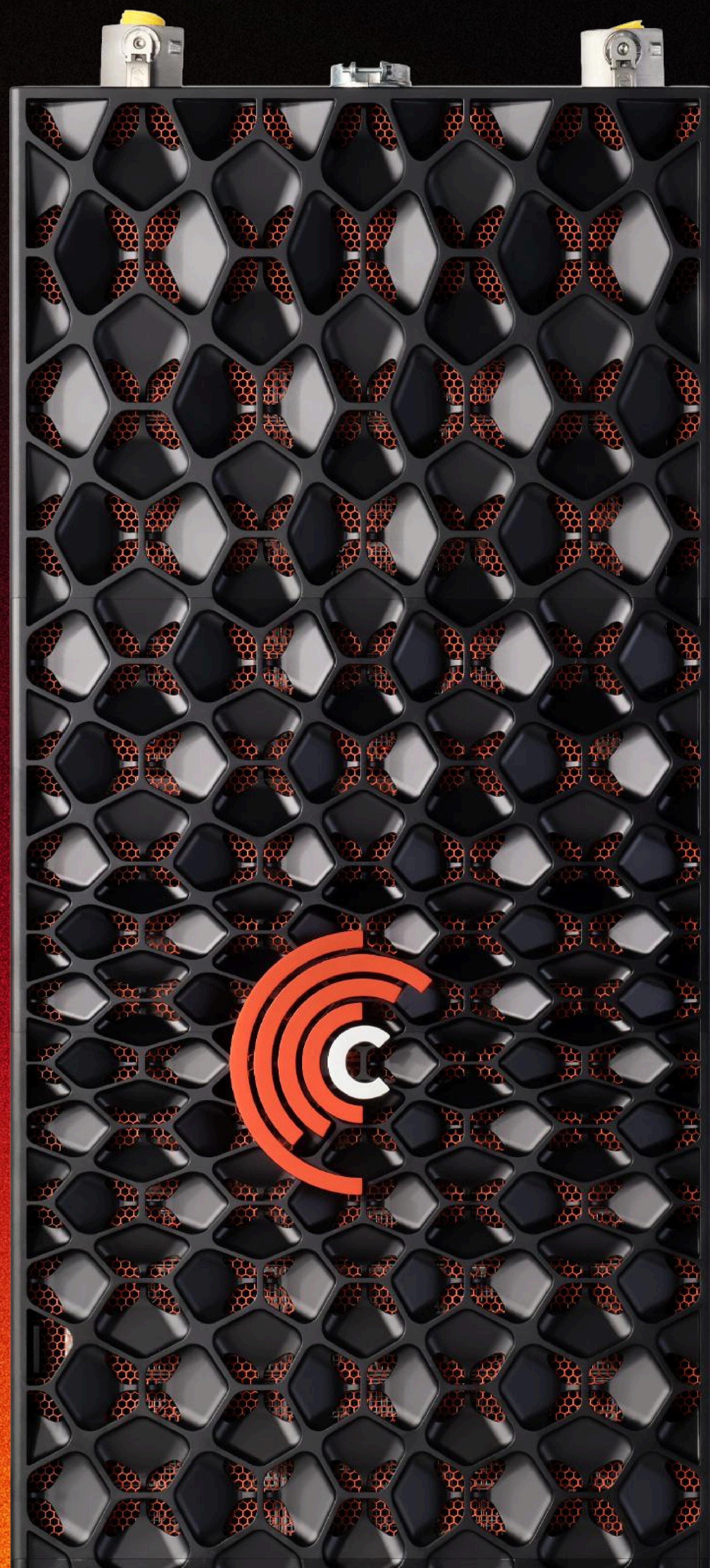


MEET CS-4

The Fastest AI Just Got Faster

Up to 30x Faster than GPU



The Fastest AI Just Got Faster.

Cerebras CS-4 is the fastest AI accelerator in the industry. The CS-4 is a rack-scale solution built from three new Wafer Scale Engine 3 Turbo processors and a revolutionary rack and system design. CS-4 delivers up to 30x faster inference compared to GPUs, better economics, and a simple path to deploy hyperscale capacity. It is the ideal architecture for frontier AI.

Up To 30X Faster than GPUs

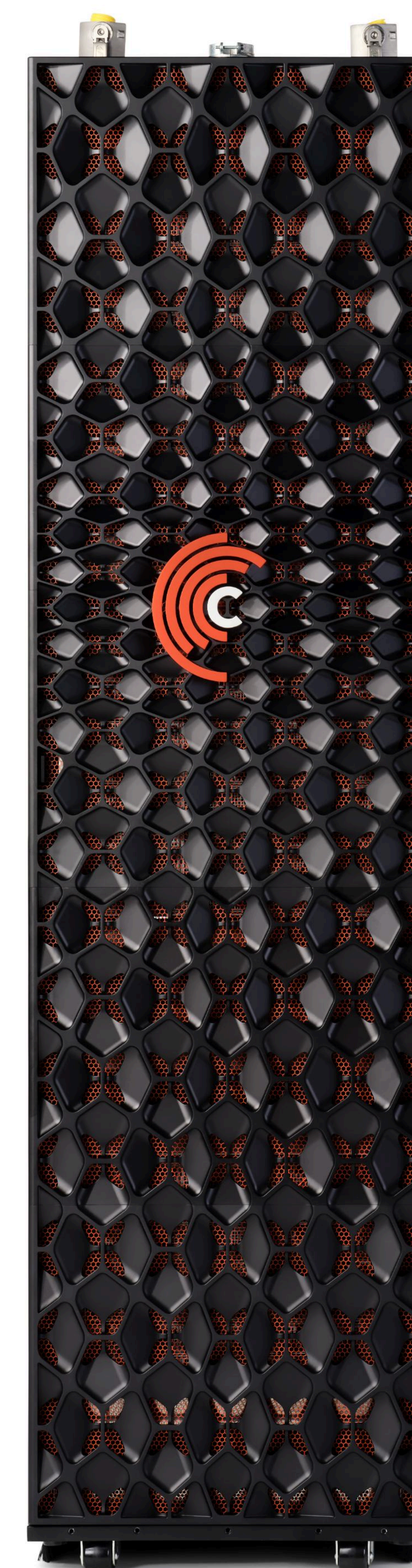
Up to 30x faster than GPUs and twice as fast as the CS-3, CS-4's more powerful processor and rack design enables a better experience for highly interactive, agentic, and reasoning workloads.

Higher Ultrafast Throughput

The CS-4 solution delivers up to 10x throughput per watt compared to the CS-3, vastly improving datacenter economics. CS-4 delivers both higher-value fast tokens and more total tokens within a given power budget – enabling higher profitability for your AI data center.

Modular Rack Design Built for Hyperscale

CS-4's modular rack design, with 50% fewer components and self-contained assemblies for compute, power, and I/O, enables faster manufacturing and rapid deployments in hyperscale datacenters. The modularity also enables independent innovation across compute, power, and I/O modules.



Engineered for Speed at Scale

WSE-3 Turbo

Delivers up to twice the speed of the WSE-3 – the new fastest in the industry

Faster Wafer I/O

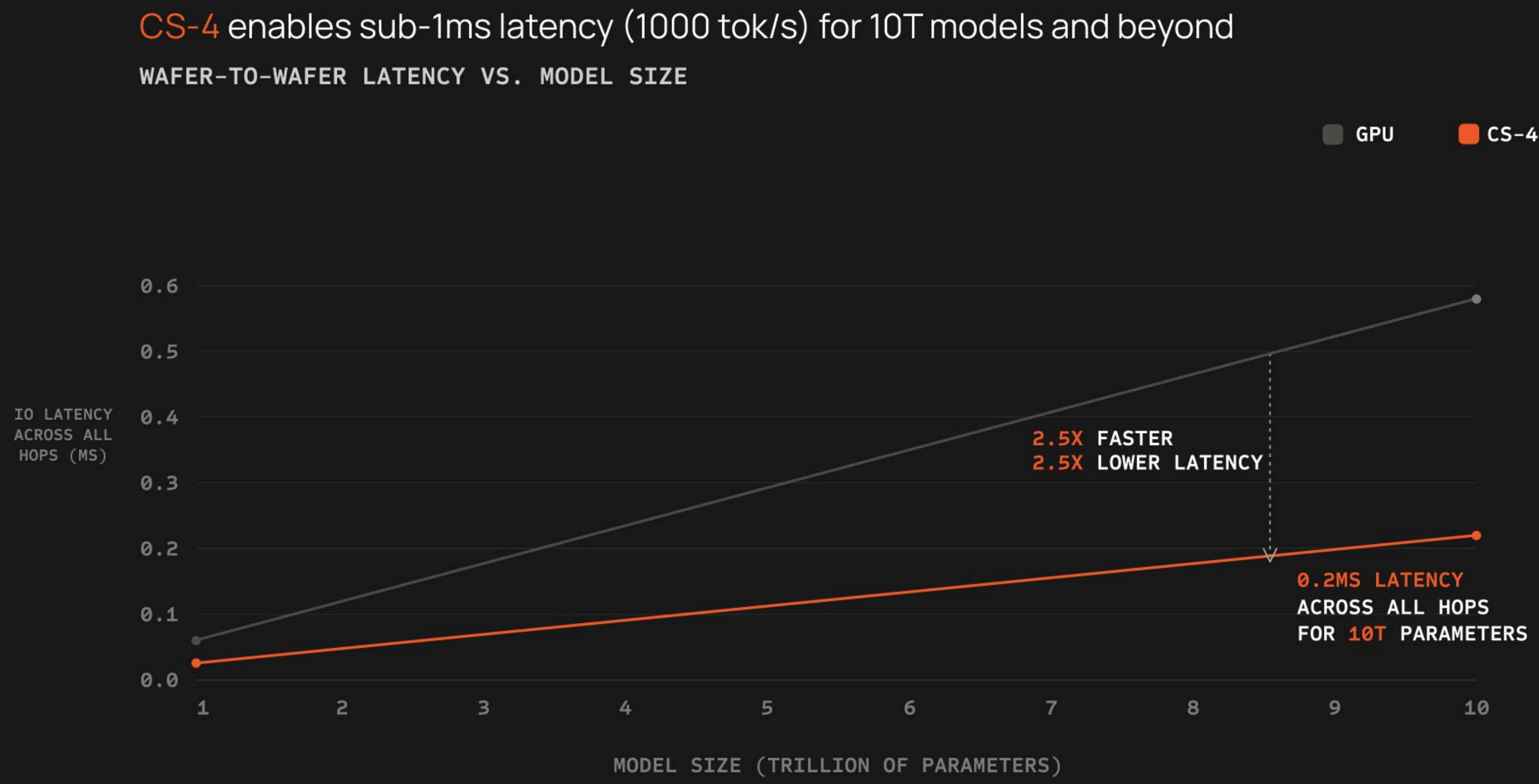
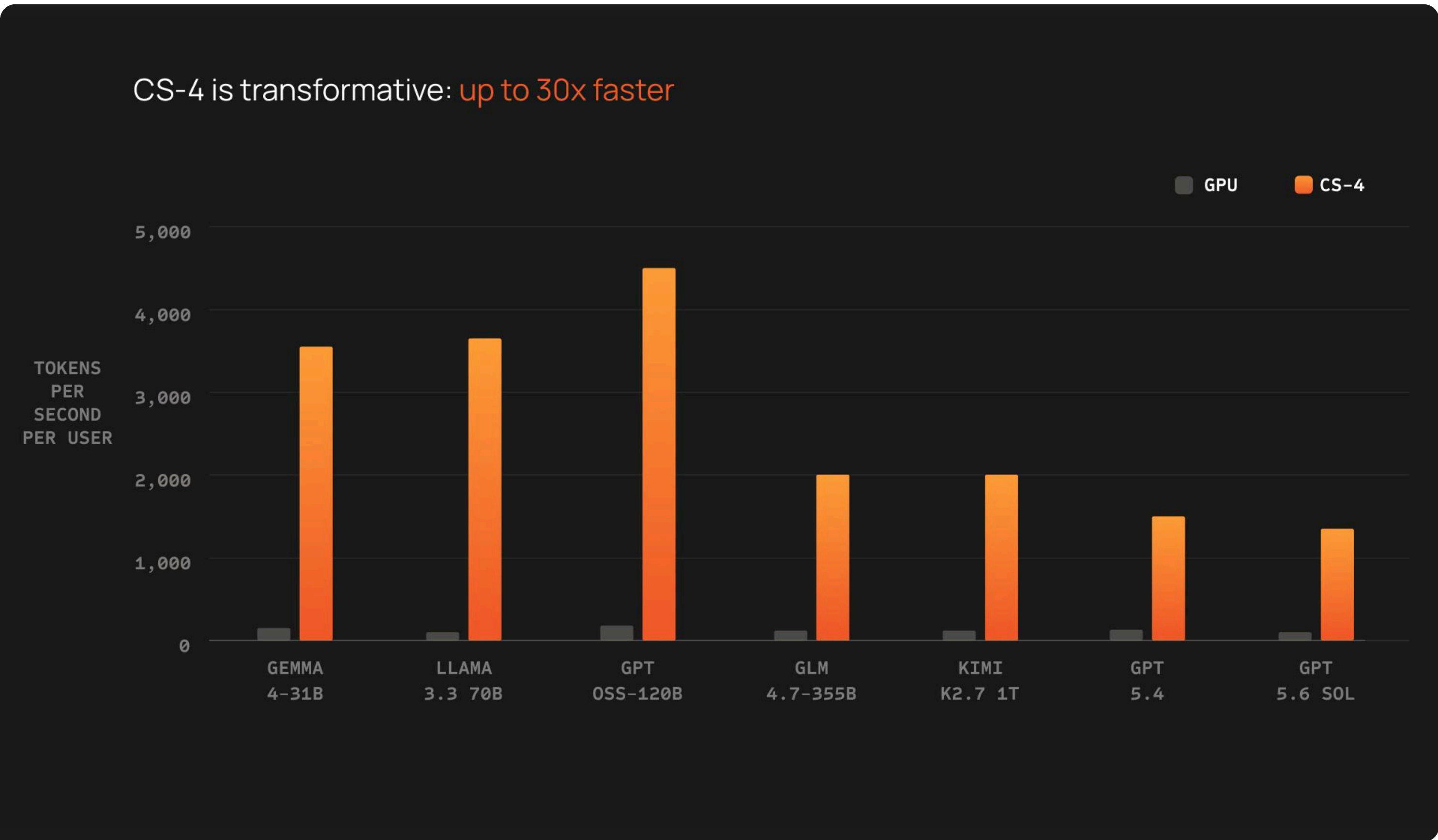
Scales massive models and enables heterogeneous, disaggregated inference

Nexus Rack-Scale Platform

Enables rapid deployment in hyperscale data centers

Up to 30X Faster Than GPUs

Powered by WSE-Turbo, CS-4 delivers up to 30x faster inference compared to GPU systems, setting a new record for the fastest inference available in production.

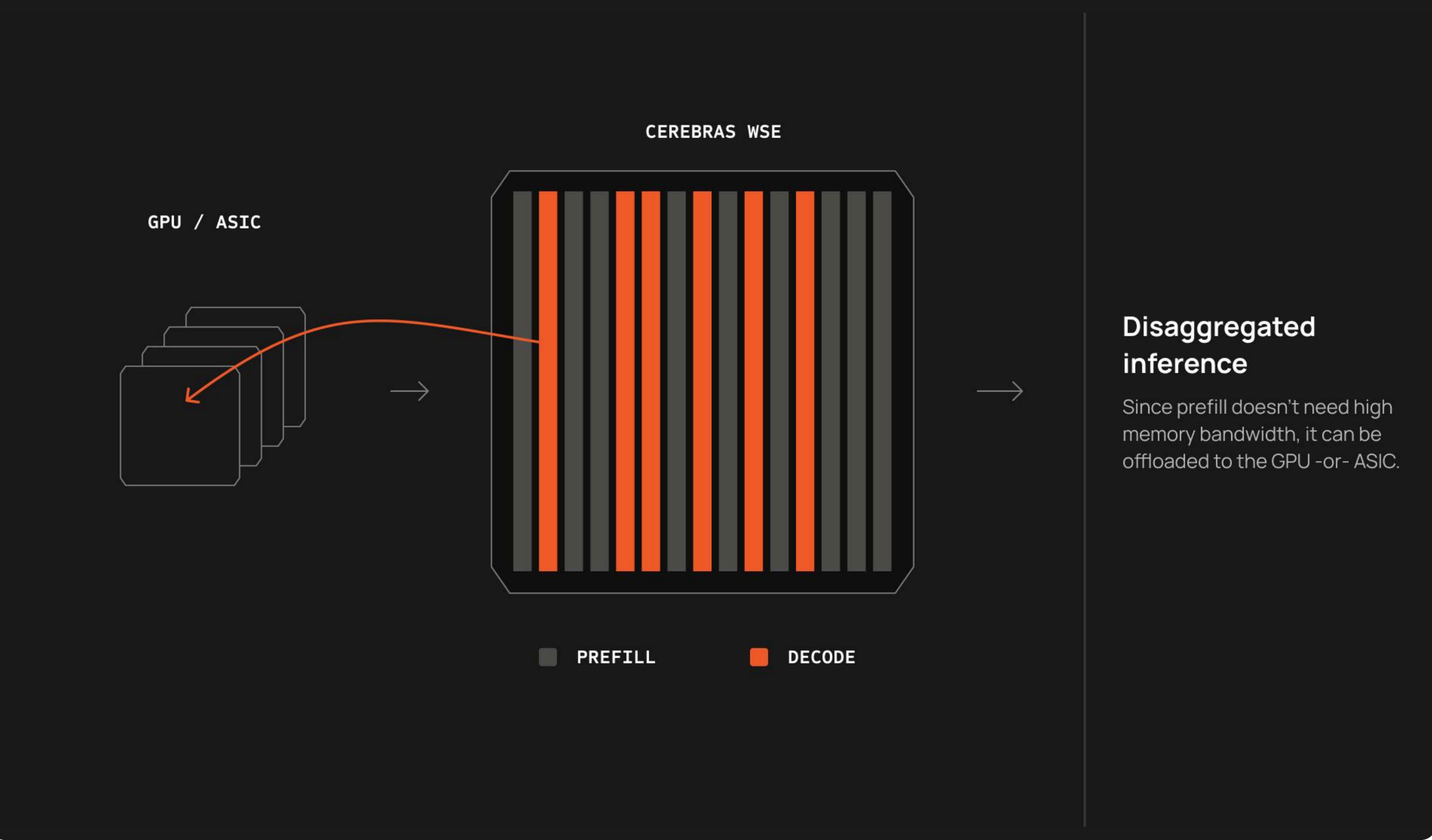
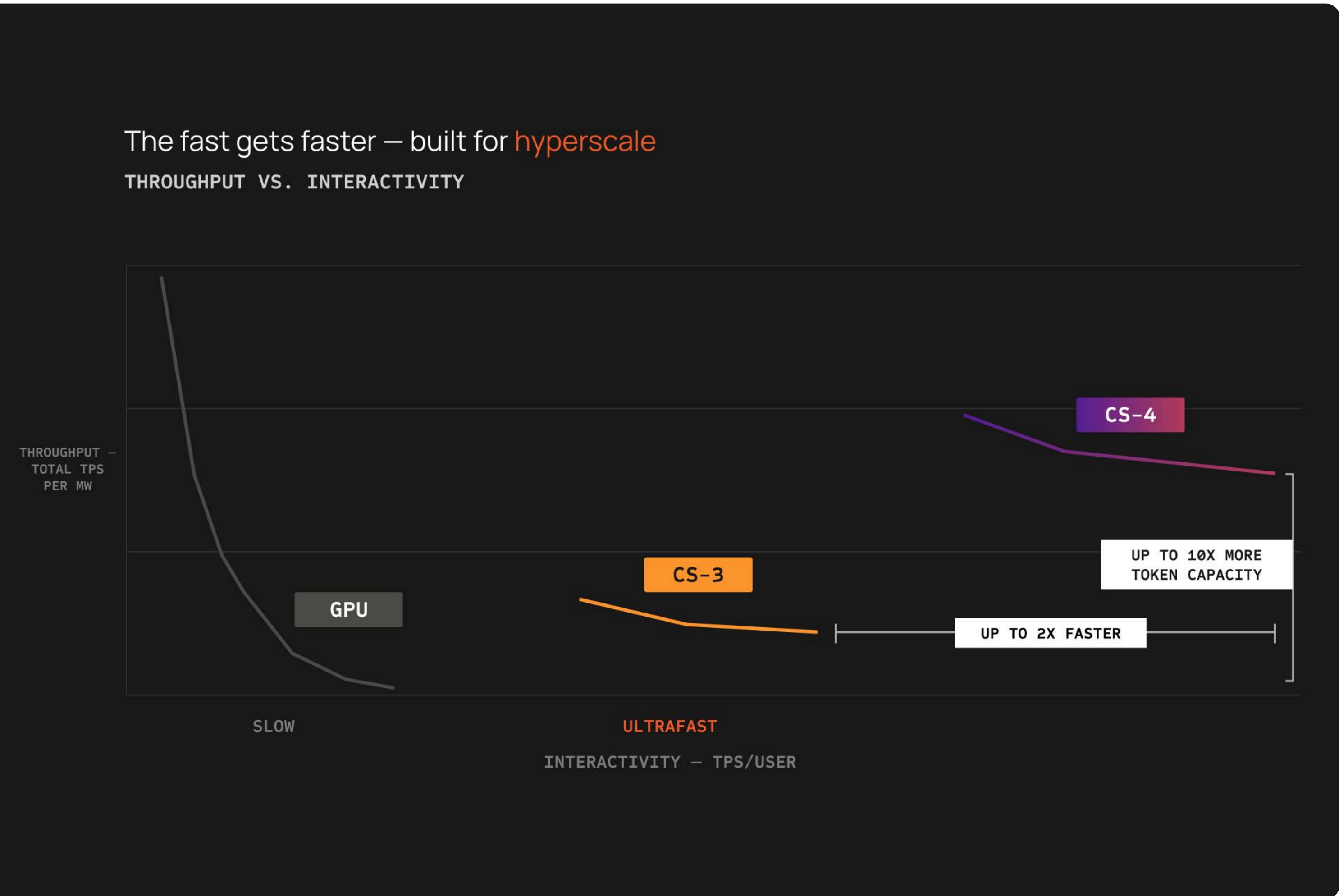


1000 Tokens per Seconds for 10T Models and Beyond

By reducing wafer-to-wafer interconnect latency to 2 microseconds, CS-4 delivers more than 1,000 tokens per second on models exceeding 10 trillion parameters, preserving interactive decode performance at unprecedented scale.

Higher Ultrafast Throughput

The CS-4 solution shifts the inference Pareto frontier, delivering up to 10x more throughput per watt than CS-3 while generating tokens up to 30x faster than production GPU systems.. The result is a system designed to deliver both throughput and interactivity. Data center operators can generate more total tokens—and more high-value fast tokens—within the same power budget, vastly improving data center economics.



Native Support for Disaggregated Inference

CS-4 pairs purpose-built prefill engines with ultra-low-latency Cerebras decoding, enabling flexible disaggregated inference across platforms including AMD Helios and AWS Trainium.

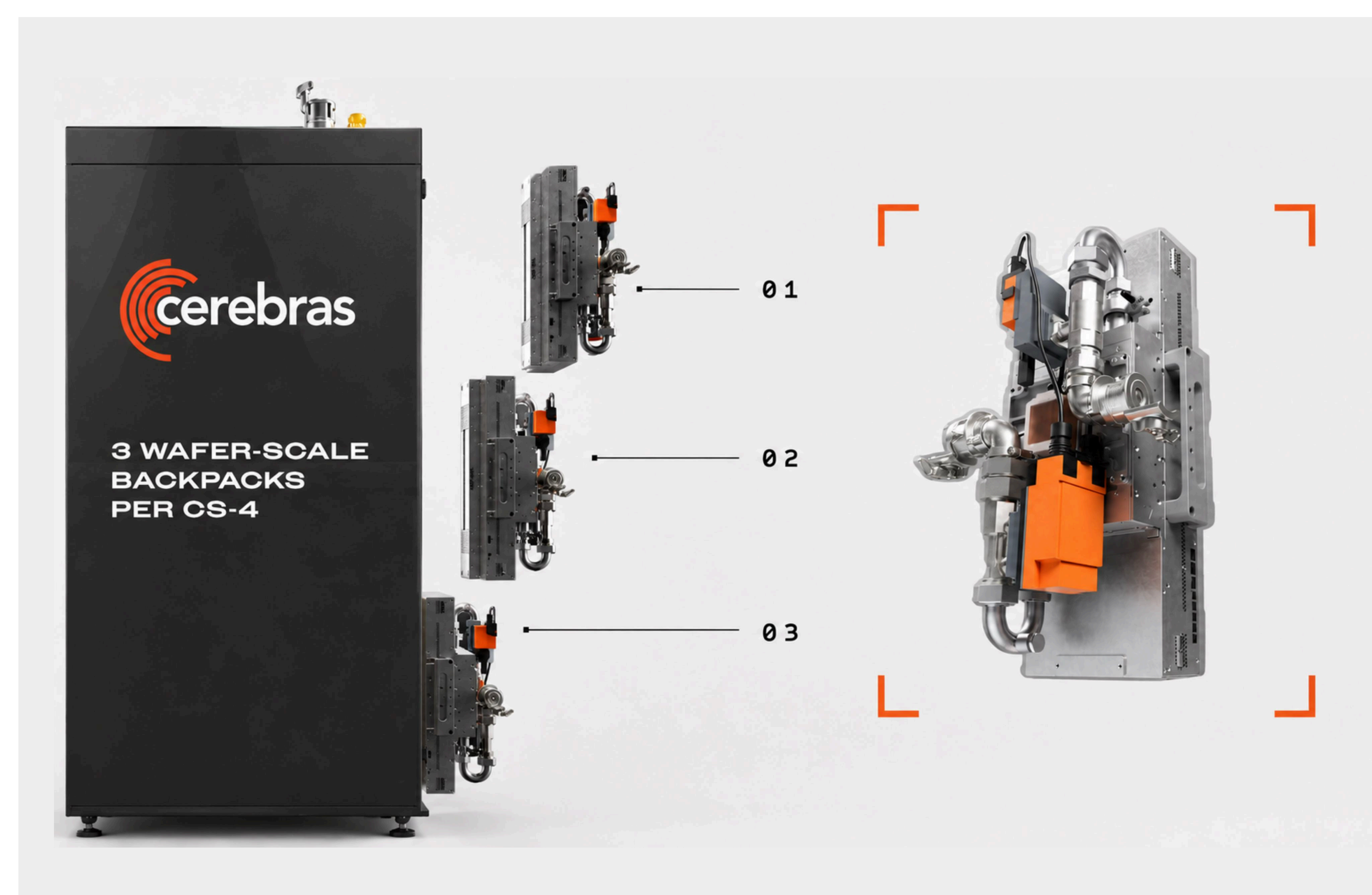
The Nexus Rack-Scale Platform – a Modular Rack Design to Enable Faster Deployments

CS-4 is the first iteration of the new Cerebras Nexus Platform Architecture. It is built around a modular concept with three foundational elements: compute, power, and I/O. The Nexus Platform brings significant innovation to each element. Modularity supports extremely rapid deployment and upgrades, as well as the ability for each element to scale independently so innovations get to market faster.



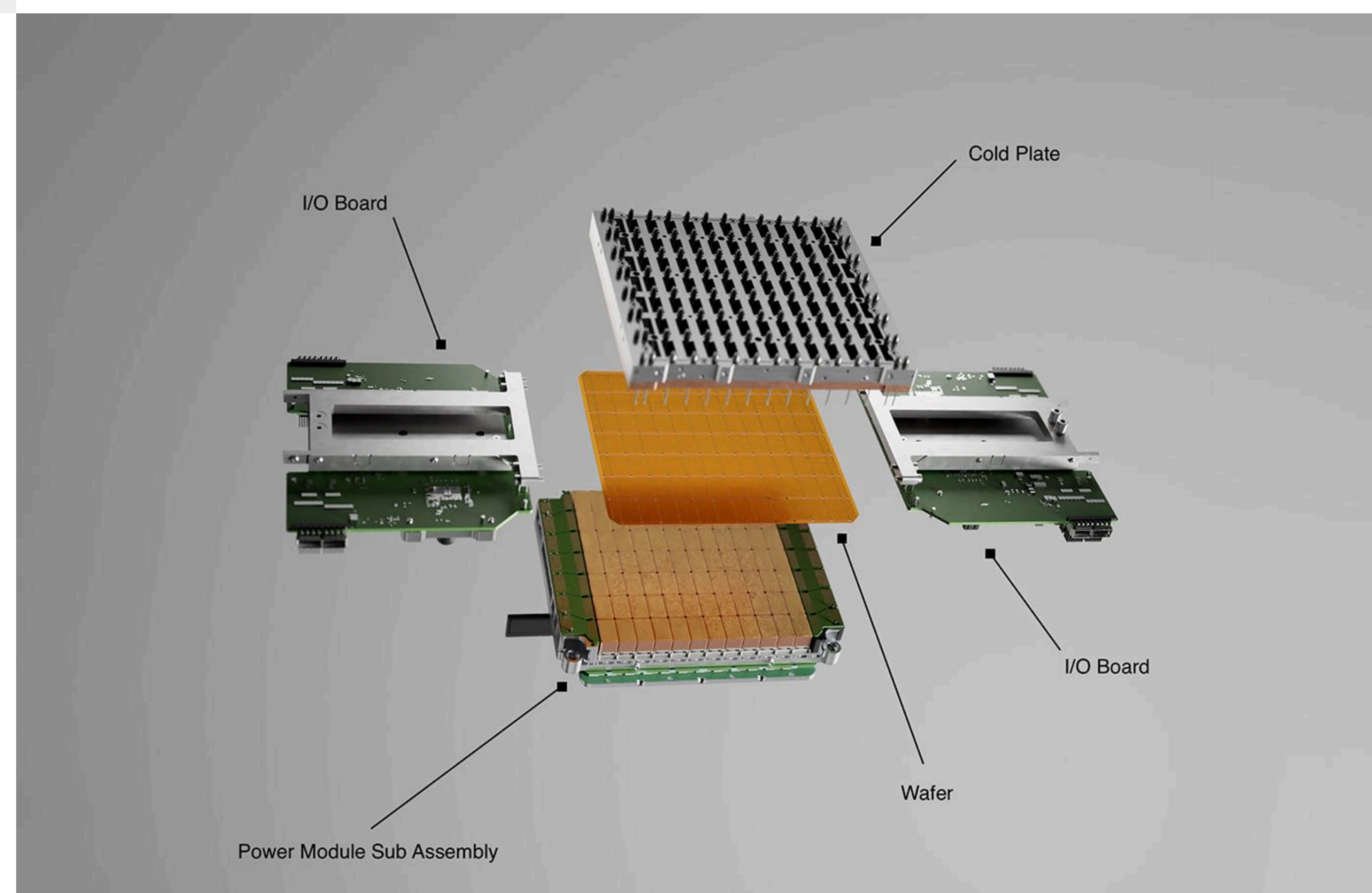
Pluggable Compute Backpack Design

Cerebras has fundamentally re-imagined the compute subsystem into a rear mounted “backpack” that attaches vertically to the power array. Each Wafer-Scale Backpack is a self-contained assembly that folds power conversion, direct liquid cooling, high-speed I/O, and control electronics into a compact, three-dimensional package built directly around the wafer. By decoupling compute from the power supplies, the Wafer-Scale Backpack simplifies manufacturing and reduces deployment time from days to hours. Compared with the prior-generation system, the Wafer-Scale Backpack has 50% fewer components and uses 60% more automated manufacturing.



High Density Power Delivery

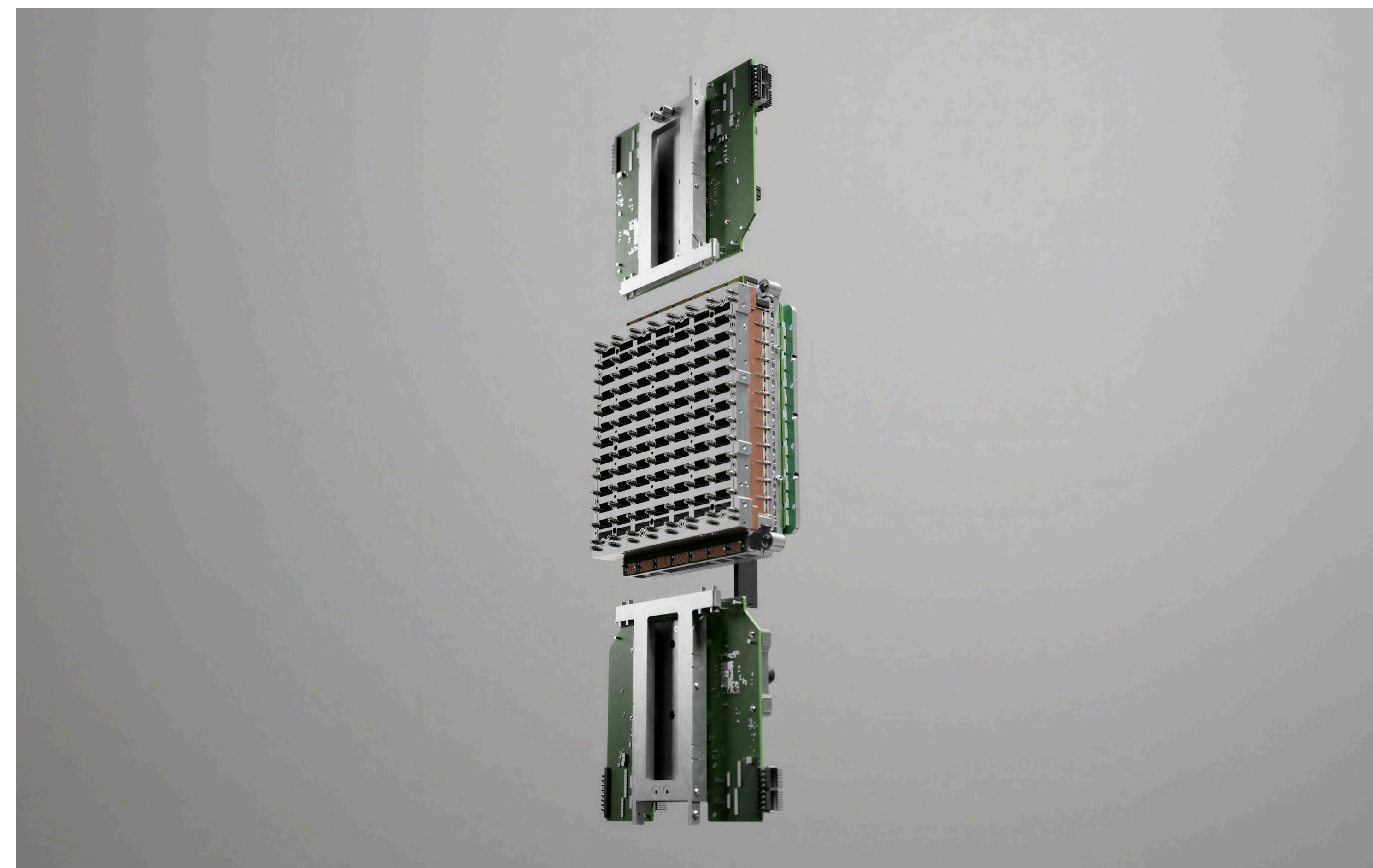
The Nexus Platform Architecture drives significant power delivery improvements. With power delivery just 0.5 millimeters away from the processor – approximately 100x closer than the roughly 50 mm of conventional GPU boards – CS-4 nearly eliminates board-level power loss. This enables the delivery of twice as much power to the WSE-3T, enabling higher operating frequencies and faster token generation.



New Modular IO Subsystem

CS-4 introduces a new programmable I/O subsystem that doubles I/O bandwidth while slashing latency.

The new Wafer I/O Module doubles I/O bandwidth and supports two connectivity modes: standards-based RoCE v2 RDMA over Ethernet for seamless integration with an ecosystem of heterogeneous systems, and Direct Wafer Links for switch-free connections within and across racks. Direct Wafer Links delivers wafer-to-wafer latency as low as 2 microseconds, enabling massive CS-4 clusters to serve frontier AI models at over 1000 tokens per second.



CS-4 Specifications

CS-4 SYSTEM

Architecture	Active power rack with three modular compute backpacks
Wafer Scale Engines	3 × WSE-3 Turbos per system
AI Compute	750 PFLOPS*
Memory Bandwidth	129.6 PB/s
On-chip Fabric Bandwidth	160.5 PB/s
I/O Bandwidth	7.2 Tbit/s
I/O latency	2 microseconds

WSE-3 Turbo (PER WAFER)

Transistors / Silicon	4 trillion transistors; 46,225 mm²; TSMC 5 nm
AI Cores / On-Chip SRAM	900,000 cores; 44 GB SRAM
AI Compute	250 PFLOPS*
Memory Bandwidth	43.2 PB/s
On-Chip Fabric Bandwidth	53.5 PB/s
I/O Bandwidth	2.4 Tbit/s

*AI compute shown as sparse FP16.